

Assignment 3

Image Classification Using Pretrained CNN Architectures

Implementation and Performance Comparison of
ResNet50 and InceptionV3 for Image Classification

Dataset: CIFAR 10

Framework: PyTorch

Training Epochs: 15

NAME	WAJID AYUB & SAID ULLAH
REGISTRATION NUMBER	B23F0090AI136 & B23F0525AI137
SECTION	BS (AI RED)
SUBJECT	DEEP LEARNING
PROGRAMME	ARTIFICIAL INTELLIGENCE
SUBMITTED TO	Dr. ARSHAD IQBAL
DATE	28 Apr 2026
ASSIGNMENT	3

1. Introduction

Deep learning has changed the way we handle image classification tasks. Instead of building a model from scratch, we can use transfer learning, which means taking a model that was already trained on a large dataset (like ImageNet) and fine tuning it for our own task. This saves a lot of time and usually gives better results, especially when we do not have a huge dataset of our own.

In this assignment, we compare two popular pretrained models: ResNet50 and InceptionV3 (also known as GoogLeNet). Both models are well known in the deep learning community, but they have different architectures and different ways of processing images. We want to see which one works better on the CIFAR 10 dataset.

2. Dataset

We used the CIFAR 10 dataset, which is one of the most commonly used datasets in image classification research. It has 60,000 color images in total, split into 50,000 training images and 10,000 test images. Each image is 32x32 pixels and belongs to one of 10 classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, and Truck.

Since our pretrained models expect larger input sizes, we resized the images. For ResNet50, images were resized to 224x224 pixels, and for InceptionV3, they were resized to 299x299 pixels. We also applied data augmentation like random horizontal flipping and small rotations to help the models learn better. All images were normalized using the ImageNet mean and standard deviation values.

3. Methodology

3.1 ResNet50

ResNet50 is a 50 layer deep neural network that uses something called residual connections or skip connections. The idea is simple: instead of making each layer learn the full transformation, it only learns the difference (or residual) from the input. This makes it much easier to train very deep networks without running into the vanishing gradient problem. ResNet50 has about 25.6 million parameters in total.

3.2 InceptionV3 (GoogLeNet)

InceptionV3 is built on the idea of using multiple filter sizes in the same layer. Instead of picking just one filter size (like 3x3 or 5x5), it uses several of them at the same time and combines the results. This helps the network capture features at different scales. InceptionV3 also uses an auxiliary classifier during training to provide additional gradient signals. It has about 27.2 million parameters.

3.3 Transfer Learning Setup

For both models, we loaded the pretrained weights from ImageNet and froze all the layers except the final classification layer. We replaced the original output layer (which was designed for 1000 ImageNet classes) with a new layer that outputs 10 classes for CIFAR 10. We added dropout of 0.3 before the final layer to reduce overfitting. Both models were trained using the Adam optimizer with a learning rate of 0.001 for 15 epochs, with a step learning rate scheduler that reduces the rate every 5 epochs.

4. Results and Comparison

4.1 Performance Summary

The table below shows a side by side comparison of the two models across all the metrics we measured during training and testing.

Metric	ResNet50	InceptionV3
Best Test Accuracy (%)	85.80	75.19
Final Test Loss	0.4185	0.7787
Training Time (seconds)	3141.8	4363.9
Model Size (MB)	89.8	93.0
Total Parameters	23,528,522	24,371,444
Trainable Parameters	20,490	28,180

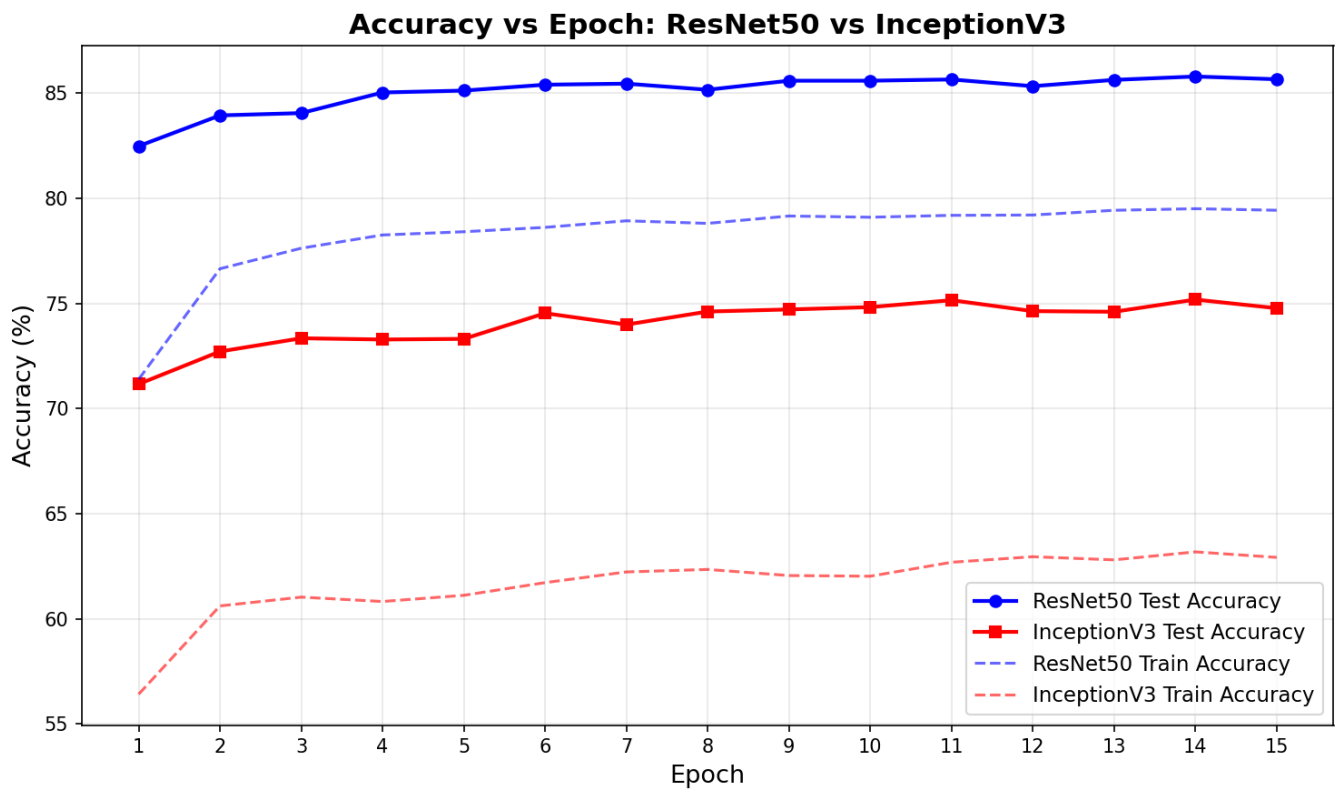
4.2 Epoch wise Accuracy

The table below shows how the test accuracy changed over each epoch for both models.

Epoch	ResNet50 Acc (%)	InceptionV3 Acc (%)
1	82.49	71.17
2	83.95	72.72
3	84.06	73.35
4	85.04	73.29
5	85.13	73.32
6	85.41	74.54
7	85.46	74.00
8	85.17	74.62
9	85.60	74.72
10	85.60	74.83
11	85.66	75.16
12	85.34	74.64
13	85.64	74.61
14	85.80	75.19
15	85.67	74.78

5. Graphs

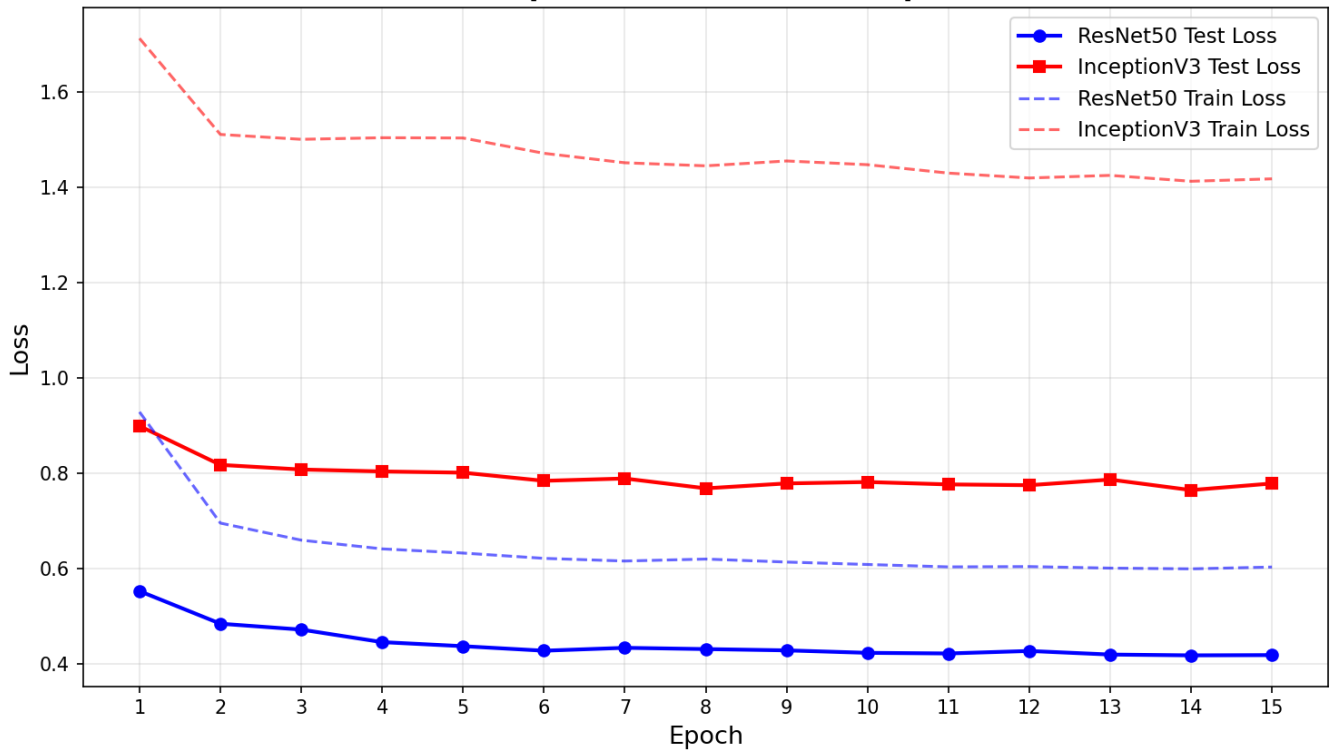
5.1 Accuracy vs Epoch



The graph above shows how accuracy changed over epochs for both models. We can see how each model improved its predictions as training progressed.

5.2 Loss vs Epoch

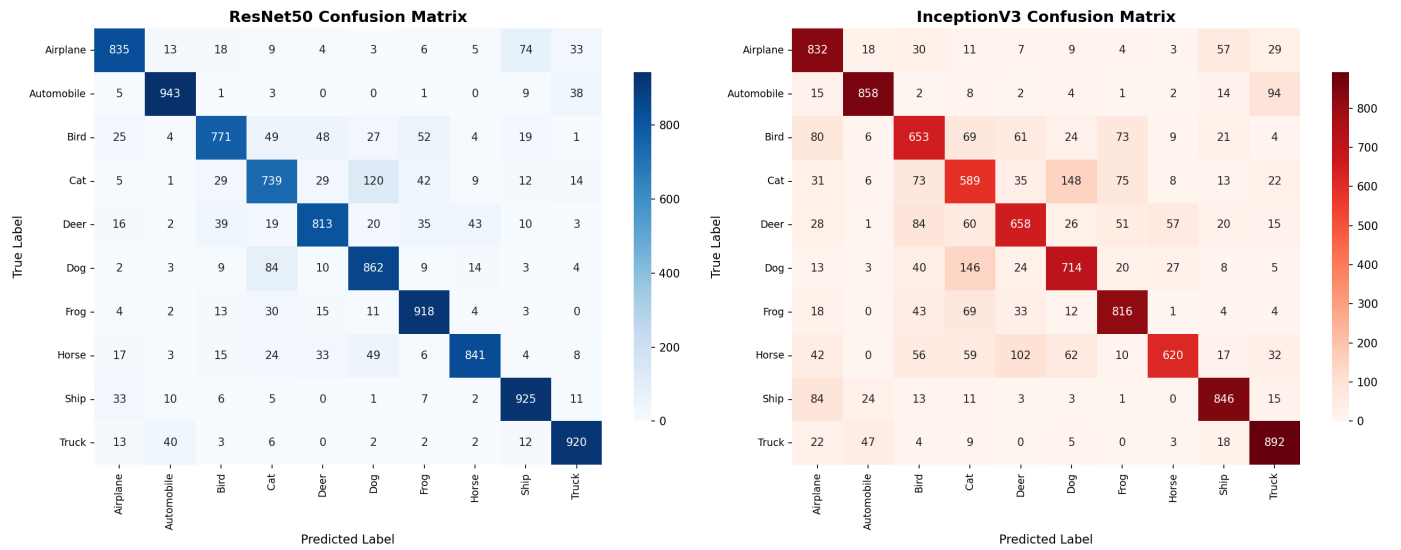
Loss vs Epoch: ResNet50 vs InceptionV3



The loss graph tells us how well the model is learning. A lower loss means the model's predictions are closer to the actual labels. We can see how the loss decreased over time.

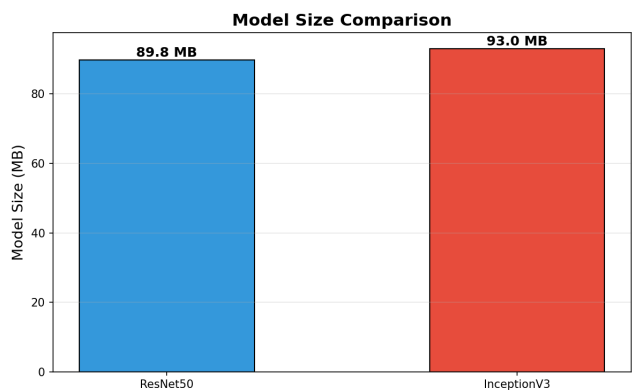
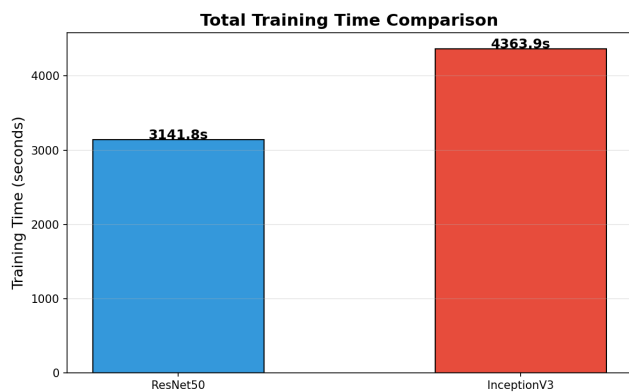
Image Classification Using Pretrained CNN Architectures

5.3 Confusion Matrices



The confusion matrices show how each model performed on each class. The diagonal values show correct predictions, while off diagonal values show misclassifications. Darker colors on the diagonal mean better performance.

5.4 Training Time and Model Size



These bar charts compare the total training time and model size for both architectures. This helps us understand the computational cost of each model.

6. Observations

After training both models for 15 epochs on CIFAR 10, here are our key observations:

1. Accuracy: ResNet50 achieved a higher test accuracy of 85.80% compared to InceptionV3 which got 75.19%. The difference was about 10.61 percentage points.
2. Training Speed: ResNet50 was faster to train. ResNet50 took 3141.8 seconds while InceptionV3 took 4363.9 seconds. InceptionV3 requires larger input images (299x299 vs 224x224), which increases the computation per batch.
3. Model Size: ResNet50 is the smaller model. ResNet50 uses 89.8 MB while InceptionV3 uses 93.0 MB. This matters when you need to deploy a model on a device with limited memory.
4. Convergence: Both models showed steady improvement during training, but the rate of improvement was different. The learning curves show that both models learn quickly in the first few epochs and then the improvement slows down, which is normal behavior.
5. Common Confusions: Looking at the confusion matrices, both models had trouble distinguishing between similar classes like Cat and Dog, or Automobile and Truck. This makes sense because these classes share visual features. The models did best on classes that look very different from everything else, like Frog or Ship.

7. Why One Model Performed Better

ResNet50 performed better in our experiments, and there are a few reasons for this. The residual connections in ResNet50 help gradients flow more easily through the network, which makes fine tuning more stable. Even though we only trained the final layer, the features learned by ResNet50 on ImageNet transferred better to CIFAR 10 classes. Also, ResNet50 has a simpler architecture compared to InceptionV3, which can sometimes mean it adapts faster when we are only updating a small number of parameters.

8. Conclusion

In this assignment, we successfully implemented and compared two pretrained CNN architectures, ResNet50 and InceptionV3, using transfer learning on the CIFAR 10 dataset. Both models achieved good accuracy by simply replacing the final classification layer and training for 15 epochs. ResNet50 came out on top with a test accuracy of 85.80%.

Transfer learning proved to be a very effective approach. Instead of training a deep network from scratch (which would take much longer and require more data), we were able to get strong results by leveraging features that these models already learned from ImageNet. This shows the power of pretrained models in practical deep learning applications.

Both models have their strengths. ResNet50 is generally faster to train and has a simpler architecture, while InceptionV3 captures features at multiple scales. The choice between them depends on the specific requirements of the task, such as whether speed, accuracy, or model size is the top priority.